NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

L26.3 PySR Wind Pressure Formula

Group Quick-Trial Lab

Status Dropped

Period 2026-10-19 17:28 — 2026-10-24 22:01

Stage conclusion

Project report

L26.3 PySR Wind Pressure Formula (Quick Trial)

Criteria (Written on 10-19 17:55, before running; unchanged afterward—any change must be logged as a decision alongside the old reading)

Data and Target Quantities - Data: All 22 cases from the TPU high-rise building (rectangular/square, isolated) aerodynamic database. Plan aspect ratios B:D = 1:1 (H/B 1–5, α=1/4, 1/6, wind direction 0–50°), 2:1 (H/short side 2–5, α=1/4, wind direction 0–100°), 3:1 (H/short side 2–5, α=1/4, 1/6, wind direction 0–100°), totaling 362 wind-direction records. - Inputs (all dimensionless): x_h = H / short side; x_r = B / D (where B is the windward width at 0°); x_a = α; wind direction is given solely by four features: cosθ, sinθ, cos2θ, sin2θ (both baselines and PySR use the exact same set). - 4 Target quantities: **Mean** and **standard deviation (fluctuating)** of the **area-averaged Cp time history** on Face 1 (windward face at 0°) and Face 3 (leeward face at 0°), respectively. The reference wind speed follows the mean wind speed at roof height in the TPU files (Cp definition unchanged).

Area averaging uses equal-weight averaging across pressure taps (if taps are non-uniform in a case, area weights are computed from TPU coordinates and noted in the results).

Holdout Sets (two types per target quantity, trained separately) - H1 Aspect-ratio extrapolation: Hold out all cases with x_h = 5 (T115_4, T115_6, T215_4, T315_4, T315_6). - H2 Wind-direction interval: Hold out θ ∈ {30°, 35°, 40°, 45°} across all cases.

Baselines (parameters and term selection determined using the training set only) - B-poly: Polynomial up to degree 2 in features {x_h, x_r, x_a, cosθ, sinθ, cos2θ, sin2θ}, forward stepwise selection, with number of terms K chosen via 5-fold cross-validation grouped by case within the training set. - B-pow: (c0 + c1 cosθ + c2 sinθ + c3 cos2θ + c4 sin2θ) · x_h^p · x_r^q · (α/0.2)^s, nonlinear least squares, 8 parameters. - "Best baseline" = the one with the smaller holdout RMSE between the two baselines. - Code ASCE 7-22 Figure 27.3-1 (Zotero EVN9AM7V, PDF page 345 / print page 284): Leeward Cp = −0.5 (L/B 0–1), −0.3 (2), −0.2 (≥4), linear interpolation; side walls −0.7; windward wall 0.8 (referenced to q_z, which differs from TPU's roof-height wind speed reference and serves only as an order-of-magnitude reference). The code applies only to 0°/90° and gives only mean values; **reported for comparison only, not part of the criteria**.

PySR Settings: pysr 2.7.2 (tag v2.7.2, commit), binary operators + − × ÷, unary operators square, exp, cos (raw θ not provided); niterations=200 per run (set on 10-19 18:15 before running, without a short pilot test; the dataset has only ~300 rows, so 200 iterations is sufficient to converge to a stable frontier—if the frontier across 5 seeds clearly fails to converge, it will be noted separately), 5 seeds; model selection from the Pareto frontier: hold out 20% of cases within the training set for validation, pick the model with the lowest validation RMSE, or the simpler model if within ≤2%.

Operational Definitions of Criterion Terms - "Non-inferior to best baseline": PySR holdout RMSE ≤ 1.05 × best baseline holdout RMSE. Also report the strict reading (≤ 1.00×) alongside. - "No more terms than baseline": Number of free constants in the PySR formula ≤ number of free parameters in the best baseline. Also report a second reading alongside: number of top-level additive terms after sympy simplification. - "Dimensionally reasonable": (1) uses only dimensionless inputs (guaranteed by construction); (2) wind direction enters only through periodic functions (guaranteed by construction); (3) finite everywhere on a dense grid over the input box x_h∈[1,6], x_r∈[1,3], α∈{1/6,1/4}, θ∈[0°,100°], with mean Cp ∈ [−2, 1.2] and fluctuating Cp ∈ [0, 1]. Failure of (3) means unreasonable. - "Isomorphic": After sympy simplification and replacing all numerical constants with placeholders, two expressions have the same canonical form (srepr). "≥3 seeds isomorphic" = at least 3 out of 5 seeds are pairwise isomorphic. Also report a relaxed reading alongside: identical set of used features and identical number of top-level additive terms. Seed stability is evaluated on the H1 training set. - **Supplement (10-19 18:10, before running, prior to seeing any data)**: PySR's holdout RMSE, constant count, and dimensional check are all evaluated on the formula from the seed with the **median holdout RMSE** among the 5 seeds; the best and worst values across the 5 seeds are reported alongside. - A target quantity "passes" = non-inferior to best baseline on both H1 and H2 holdouts + no more terms than best baseline on both holdouts + dimensionally reasonable + ≥3 seeds isomorphic (all four conditions met). - **Worth using** = ≥2 out of the 4 target quantities pass; otherwise **Abandon**.

Data Source and Terms of Use (Verified 10-19 17:35)

- Official website [external URL omitted] : publicly downloadable, requires citing "TPU (Tokyo Polytechnic University) Aerodynamic Database, 2026, [external URL omitted] all rights reserved, no open license. Handling: internal research use only, no redistribution of raw data; public pages include only summary statistics, formulas, and plots.
- Our center does not have a complete local copy (checked TPU: R25 is Not started, not downloaded; L29.1 only downloaded the 0° files for T114_6 and T115_4).
- Precedent: Wang et al. 2025, J. Build. Eng. 99:111614 (card, PDF pp.4–17) applied GP symbolic regression on the TPU **interference effects** database to fit mean overall force coefficients; used a random 70/30 split within the same database, without extrapolation holdouts or seed stability checks. Extrapolation and stability are precisely what this quick trial addresses.

Security Check (PySR, 10-19 17:45, commit shallow-cloned in working directory, read-only)

- License: Apache-2.0. Installation uses pyproject (hatchling); setup.py only prints a message and exits, with no post-install scripts.
- Outbound connections: First run downloads Julia 1.10 and SymbolicRegression.jl ~2.7.2 from official Julia sources (written into the virtual environment directory). No other network calls in the source code (grep urllib/requests/http/socket: only a local socketpair in interrupt.py for Ctrl-C).
- `jl.seval` executes only fixed Julia strings within the package and operator names supplied by us; does not load external pickles. Does not read credentials, modify/delete user files, escalate privileges, or contain obfuscated code.
- Conclusion: Low risk. Install only in a disposable virtual environment on CRC, pin pysr==2.7.2, and do not include in any production bot.

Trials and Results

Time	Action Taken	Result	Evidence
10-19 18:40	Resource request (ComputeOps reply: J1 long/smp 8, J2 hpc/mpi-48 single node, Julia precompilation inside job, exclude d24cepyc154); installed pysr 2.7.2 / juliacall 0.9.36 (449 MB) in CRC venv; preflight qsub -w v passed; submitted J1 1592459 (statistics) and J2 1592460 (hold J1; baselines + 40 PySR runs + evaluation), both registered	Queued	CRC [local path omitted][user] [local path omitted], run_j2; script (sha256 matches CRC copy)
10-19 19:30	J1 1592459 finished normally (exit code 0, submitted at 17:40, ~1 h 50 min including queue time)	362/362 files downloaded and computed, 0 failures; compute nodes have outbound internet access. T214_4 and T215_4 have non-uniform tap layouts, weighted by area (max difference from equal weighting: 0.029). 0° windward mean Cp 0.48–0.78, leeward −0.39 to −0.75, physically reasonable order of magnitude	(contains only summary statistics and raw file sha256 hashes, no raw time series)

Time	Action Taken	Result	Evidence
10-19 19:45	J2 1592460 exited with code 0 but **invalid**: baselines completed, but all 40 PySR runs failed to start—codegen in Julia 1.13.2 (actual version installed by juliapkg, not 1.10 as written in juliapkg.json) requires GLIBCXX_3.4.30, whereas the Python process preloaded the older system libstdc++. The automatic "0/4 → Abandon" output from analyze was due to missing data rather than actual results and is voided (analyze has been updated not to evaluate incomplete results). Fix: LD_PRELOAD Julia's bundled libstdc++ in the job script, and change warmup to run 1 actual small fitting iteration; new run root run_j2b, job 1593122 (under the same allocation request), registered. run_j2 retained as-is	Running	CRC; (sha256 …)
10-19 19:49	J2b 1593122 finished normally (exit code 0, 19:35–19:49)	All 40/40 PySR runs produced result files, no errors in logs; per agreement while awaiting launch plan approval, **logged only, evaluation unread, conclusion not reported**	(424 KB, excluding PySR temp directories; original run_j2b retained on CRC)
10-21	Liang Hu approved A139 (approved plan, retroactively approved J1/J2/J2b, added one 3000-iteration long search ≤100 core-hours); pplan check cleared; reused allocation request; experiment card; submitted J3 1602747 (hpc single node 48 cores, ≤2 h, new run root run_j3, run_j2b untouched), registered with --card	Running	CRC [local path omitted][user] [local path omitted]

Time	Action Taken	Result	Evidence
10-21 15:36	J3 1602747 finished normally, 40/40	Criteria 0/4 → Abandon; accuracy improved substantially but structure remained unstable; Lead manually verified RMSE of two selected formulas, matching result files	;

Parameter Source Table

Parameter	Value	Source / Assumption
Case geometry, wind direction, α	See Criteria	TPU individual case pages (read 10-19 17:35)
Code Cp	0.8 / −0.5,−0.3,−0.2 / −0.7	ASCE 7-22 Figure 27.3-1, PDF page 345
Reasonable range: mean Cp [−2,1.2], fluctuating [0,1]	—	Assumption: physical order of magnitude for face-averaged values on isolated rectangular prisms, serving as an anti-divergence check without affecting accuracy criteria

Gate Deviation (10-19 19:40)

The master control task brief stated "Liang Hu's verbatim words constitute approval; no need to wait for the launch plan," based on which I submitted J1/J2/J2b; at 19:39 the platform reminded that verbal naming counts only as plan_only. Launch plan has been retroactively submitted pending Liang Hu's approval. J2b was nearly finished and was not force-killed; no further jobs will be submitted before approval, and J2b results are only logged without reading the evaluation or reporting conclusions.

Environment Lessons for Running PySR on CRC

Julia 1.13.2 (installed by juliapkg) requires GLIBCXX_3.4.30; in the job script, export LD_PRELOAD=<venv>[local path omitted]++.so.6, and run a real fitting warmup with niterations=1, exiting immediately on failure.

Open Items and Next Steps

- Resource request (computeops) → install environment on CRC, download and compute statistics (delete raw files immediately after processing) → baselines → PySR short pilot test to set niterations → formal 40 runs → evaluation.

Citation Snowballing Expansion (10-22, Liang Hu directive to re-run across all sub-projects)

- Execution and screening: , screening log.
- Card creation delegated to litwiki (conclusions and criteria remain unchanged until cards return).
- Preliminary impact: The verdict "Abandon (limited to this configuration)" remains unchanged; if benchmark literature reports that cross-seed structural consistency of PySR is generally low, add a layer to the conclusion noting "this is a common methodological limitation."
- Card verification (10-22): Wu et al. 2026 (JWEIA, PySR and PhySO for across-wind displacement) used only an 80/20 split with fixed seed 42 and 5-fold R^2, **without reporting formula structural consistency across seeds**, nor holding out by experimental source or extrapolation regime (PDF pp. 6–7, 9, 15); SRBench++ reported ground-truth symbolic recovery rates on synthetic datasets (10 seeds), but **did not** include a statistic for cross-seed structural consistency (accepted manuscript PDF pp. 2–3, 5–6). → This trial's check for "cross-seed non-isomorphism" has not been performed in recent comparable wind engineering studies or benchmarks; conclusion unchanged, and a note that "existing literature has not examined this aspect" can be added to the handoff brief.

Key figures